Homework 4: solutions

Continuous laws, moments, and joint densities

1. A density and its CDF

5 points

The density of X is $f_X(x) = c(2-x)I_{\{0<x<2\}}$, where c is an unknown constant.

(a) [3] Find the value of c that makes f_X a valid density. Derive $F_X(t)$ for all real t, including the values at $t = 0$ and $t = 2$.

(b) [2] Use your CDF to find $P(1/2 < X \leq 3/2)$ and $P(X = 1)$. Explain the point probability.

Solution (a). Nonnegativity requires $c \geq 0$, and $1 = c\int_0^2 (2-x)\,dx = 2c$, so $c = 1/2$. Thus

$$F_X(t) = \begin{cases} 0, & t \leq 0, \\ t - t^2/4, & 0 < t < 2, \\ 1, & t \geq 2. \end{cases}$$

The interior expression approaches 0 at 0 and 1 at 2, so the branches join continuously.

Solution (b). $F_X(3/2) - F_X(1/2) = 15/16 - 7/16 = 1/2$. Also $P(X = 1) = F_X(1) - F_X(1^-) = 0$: the CDF has no jump, and a singleton has zero area under a density.

2. A CDF and its density

5 points

A random variable V has CDF

$$F_V(t) = \begin{cases} 0, & t \le 0, \\ t^2/4, & 0 < t < 2, \\ 1, & t \ge 2. \end{cases}$$

(a) [2] Find a density $f_V(v)$, including its domain, and verify that its integral over the real line is 1.

(b) [3] Calculate $f_V(3/2)$, $F_V(3/2)$, and $P(V = 3/2)$. Explain the meaning of each quantity and why a density value is not a point probability.

Solution (a). Differentiating on the open support gives $f_V(v) = (v/2)I_{\{0<v<2\}}$. The CDF is continuous at the endpoints, so there are no endpoint atoms. The normalization check is $\int_0^2 v/2\,dv = [v^2/4]_0^2 = 1$. Density values at the two endpoints can be chosen arbitrarily.

Solution (b). The three values are 3/4, 9/16, and 0. The first is a density height (probability per unit of v), the second is the probability of $V \le 3/2$, and the third is a point probability. Density is integrated over an interval; it is not itself the probability at that point.

3. Moments and a cost measurement

5 points

The density of W is $f_W(w) = 3w^2 I_{\{0<w<1\}}$. Define a cost $C = 2 + 4W$, measured in dollars.

(a) [3] Compute $E[W]$ and $E[W^2]$ by integration, and use them to find $\mathrm{Var}(W)$.

(b) [2] Find $E[C]$, $\mathrm{Var}(C)$, and $\mathrm{SD}(C)$, with units. Explain how the coefficient 4 affects the variance.

Solution (a). $E[W] = 3\int_0^1 w^3\,dw = 3/4$ and $E[W^2] = 3\int_0^1 w^4\,dw = 3/5$. Hence $\mathrm{Var}(W) = 3/5 - 9/16 = 3/80$. The second moment and square of the first moment are different quantities.

Solution (b). $E[C] = 2+4(3/4) = 5$ dollars, $\mathrm{Var}(C) = 4^2(3/80) = 3/5$ dollars squared, and $\mathrm{SD}(C) = \sqrt{3/5}$ dollars. Since $C - E[C] = 4(W - E[W])$, squaring the centered deviation introduces 4^2.

4. A uniform waiting time

5 points

A waiting time U, measured in minutes, is uniform on $(2, 8)$.

(a) [2] Write the density of U and state where it is zero. Find $E[U]$ and $\mathrm{Var}(U)$, with units.

(b) [3] Calculate $P(U > 5)$, $P(U > 7 \mid U > 4)$, and $P(U > 3)$. Use the last two probabilities to decide whether this distribution has the memoryless property. Explain their meaning in terms of a remaining wait.

Solution (a). $f_U(u) = \frac{1}{6} I_{\{2<u<8\}}$. The midpoint is $E[U] = (2 + 8)/2 = 5$ minutes, and $\mathrm{Var}(U) = (8 - 2)^2/12 = 3$ minutes squared.

Solution (b). $P(U > 5) = 3/6 = 1/2$. Since $\{U > 7\} \subset \{U > 4\}$,

$$P(U > 7 \mid U > 4) = \frac{1/6}{4/6} = \frac{1}{4}.$$

But $P(U > 3) = 5/6$. Thus the conditional chance of waiting three more minutes after four minutes is not the chance of an initial wait exceeding three minutes; the law is not memoryless.

5. An exponential processing time

5 points

A processing time T is exponential with mean 5 minutes.

(a) [3] Find the rate λ, with units, and write the density of T. Calculate $P(T > 3)$ and $\mathrm{Var}(T)$, with units for the variance.

(b) [2] Calculate $P(T > 9 \mid T > 4)$ from the definition of conditional probability. Interpret the event as a remaining wait after four minutes have elapsed.

Solution (a). The rate is $\lambda = 1/5$ per minute, so $f_T(t) = \frac{1}{5}e^{-t/5}I_{\{t>0\}}$. Integration gives $P(T > 3) = \int_3^\infty \frac{1}{5}e^{-t/5}\,dt = e^{-3/5}$. The variance is $1/\lambda^2 = 25$ minutes squared.

Solution (b). Because $\{T > 9\} \subset \{T > 4\}$ and $P(T > 4) > 0$,

$$P(T > 9 \mid T > 4) = \frac{e^{-9/5}}{e^{-4/5}} = e^{-1} = P(T > 5).$$

Under this exponential model, after four minutes the chance of at least five additional minutes equals the chance of an initial wait exceeding five minutes.

6. Probabilities under a normal model

5 points

A measurement X, in millivolts, has distribution $N(80, 36)$. Use $\Phi(1) = 0.841345$ and $\Phi(2) = 0.977250$.

(a) [3] Write the density of X and state its domain. Find $P(74 < X < 92)$, showing the standardized endpoints.

(b) [2] Find $P(|X - 80| > 12)$. Explain which measure of variability belongs in the denominator when standardizing X.

Solution (a). The SD is 6 mV and the support is all of $\mathbb{R}$. Thus

$$f_X(x) = \frac{1}{6\sqrt{2\pi}} \exp\left(-\frac{(x-80)^2}{72}\right), \quad x \in \mathbb{R}.$$

The endpoints become -1 and 2, so the probability is $\Phi(2) - \Phi(-1) = \Phi(2) + \Phi(1) - 1 = 0.818595$. An all-real support indicator would add no information.

Solution (b). $P(|X - 80| > 12) = P(|Z| > 2) = 2[1 - \Phi(2)] = 0.045500$. Dividing by the SD makes $Z = (X - 80)/6$ unitless with variance one; dividing by the variance does not.

7. A joint density on a triangle

10 points

For Questions 7 and 8, X and Y have joint density $f_{X,Y}(x,y) = cI_{\{0<x<y<2\}}$, where c is an unknown constant.

(a) [5] Sketch the region where the density is nonzero and label its boundary lines. Find c by integration. Write the normalization integral in both orders: $dy\,dx$ and $dx\,dy$.

(b) [5] Find $f_X(x)$ and $f_Y(y)$, including their domains. Verify that each density integrates to 1. Explain how the joint-density region accounts for the shapes of the two marginal densities.

Solution (a). The support is the triangle above $y = x$ and below $y = 2$, with $x > 0$. Its area is 2. In either order,

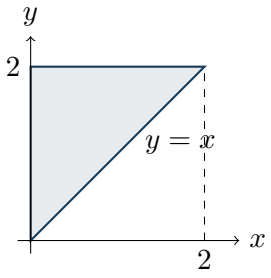
$$1 = \int_0^2 \int_x^2 c\,dy\,dx = \int_0^2 \int_0^y c\,dx\,dy = 2c,$$

so $c = 1/2$. The boundary lines have probability zero.

Solution (b). For $0 < x < 2$, $\int_x^2 (1/2)\,dy = 1 - x/2$; for $0 < y < 2$, $\int_0^y (1/2)\,dx = y/2$. Therefore

$$f_X(x) = (1 - x/2)I_{\{0<x<2\}}, \qquad f_Y(y) = (y/2)I_{\{0<y<2\}}.$$

Each integral is one: $[x - x^2/4]_0^2 = 1$ and $[y^2/4]_0^2 = 1$. Larger starts allow shorter vertical slices; larger finishes allow longer horizontal slices.



8. Event probabilities from a joint density

5 points

Use the joint density from Question 7.

(a) [2] Calculate $P(X > 1, Y < 1)$ and $P(X > 1)P(Y < 1)$. Use the comparison to decide whether X and Y are independent.

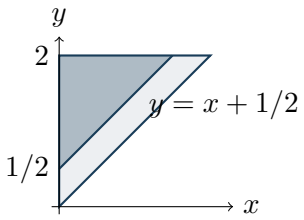
(b) [3] Sketch the part of the joint-density region where $Y - X > 1/2$. Write a double integral for its probability in each order, $dy\,dx$ and $dx\,dy$, and evaluate either one.

Solution (a). The joint event is impossible because the support requires $X < Y$, so $P(X > 1, Y < 1) = 0$. From the marginals, $P(X > 1) = \int_1^2 (1 - x/2)\,dx = 1/4$ and $P(Y < 1) = \int_0^1 y/2\,dy = 1/4$. Their product is $1/16 \neq 0$, so X and Y are dependent.

Solution (b). The event is the triangle $0 < x < 3/2$, $x + 1/2 < y < 2$. Hence

$$P(Y - X > 1/2) = \int_0^{3/2} \int_{x+1/2}^{2} \frac{1}{2}\,dy\,dx = \int_{1/2}^{2} \int_0^{y-1/2} \frac{1}{2}\,dx\,dy = \frac{9}{16}.$$

For example, the first integral is $\frac{1}{2}\int_0^{3/2}(3/2 - x)\,dx = 9/16$.



9. A joint density on a square

5 points

The joint density of R and S is $f_{R,S}(r,s) = 2r I_{\{0<r<1,\ 0<s<1\}}$.

(a) [3] Find $f_R(r)$ and $f_S(s)$, including their domains, and verify that each integrates to 1.

(b) [2] Find $P(S < R)$. Would the fraction of the square lying below the line $s = r$ give this probability? Justify your answer.

Solution (a). $f_R(r) = \int_0^1 2r\,ds = 2r$ for $0 < r < 1$, and $f_S(s) = \int_0^1 2r\,dr = 1$ for $0 < s < 1$, with zero elsewhere. Thus

$$f_R(r) = 2r I_{\{0<r<1\}}, \qquad f_S(s) = I_{\{0<s<1\}}.$$

The checks are $\int_0^1 2r\,dr = 1$ and $\int_0^1 1\,ds = 1$.

Solution (b). $P(S < R) = \int_0^1 \int_0^r 2r\,ds\,dr = \int_0^1 2r^2\,dr = 2/3$. An area ratio applies to a constant joint density. Here the density grows with r and gives more weight to the part of the square where $s < r$.

10. A service deadline

10 points

A job has two independent stage durations: $A \sim \text{Uniform}(0, 2)$ and $B \sim \text{Uniform}(0, 4)$, both measured in seconds. It misses a four-second deadline when $A + B > 4$.

(a) [5] Write the joint density of A and B, including its domain. Sketch the part of that domain where the job misses the deadline, and calculate $P(A + B > 4)$.

(b) [5] Calculate $P(A + B > 4 \mid A > 1)$. Compare it with part (a), and explain the effect of knowing that the first stage lasts more than one second.

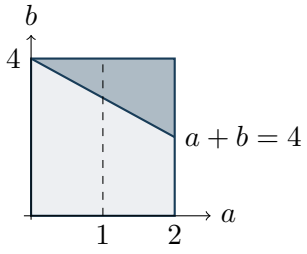
Solution (a). The rectangle has area 8, so normalization gives $f_{A,B}(a, b) = \frac{1}{8} I_{\{0<a<2,\ 0<b<4\}}$. Within this rectangle the missed-deadline region satisfies $4 - a < b < 4$. It is a triangle with vertices $(0, 4), (2, 2), (2, 4)$. Thus

$$P(A + B > 4) = \int_0^2 \int_{4-a}^4 \frac{1}{8}\, db\, da = \frac{1}{8} \int_0^2 a\, da = \frac{1}{4}.$$

Solution (b). $P(A > 1) = 1/2$, while

$$P(A + B > 4, A > 1) = \int_1^2 \int_{4-a}^4 \frac{1}{8}\, db\, da = \frac{3}{16}.$$

The conditional probability is $(3/16)/(1/2) = 3/8$. Conditioning retains longer first-stage durations, leaving less time for the second stage. Independence of the stages does not make the missed-deadline event independent of A.

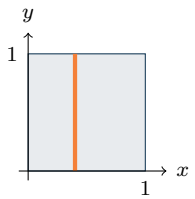


Optional practice: worked answers

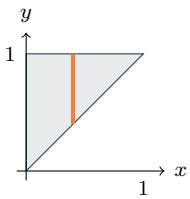
Support, normalization, and marginals

Ungraded: no credit and no submission required. This appendix is not Question 11, is not part of the rubric, and has no Gradescope entry. The graded assignment remains 10 questions and 60 points.

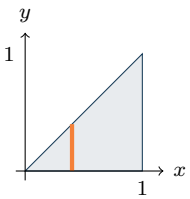
Read the regions first. A is the unit square; B is above $y = x$; C is below $y = x$; D is below $y = 1 - x$. In E, $x - y < 1$ holds throughout the open unit square, while $x - y > 0$ means $y < x$. Thus **E is the same model as C**.



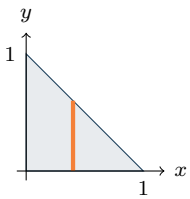
A: square



B: $x < y$



C and E: $y < x$



D: $x + y < 1$

The thick vertical segment in each sketch is one slice at fixed x. Boundary lines have probability zero, so their inclusion does not change the results.

Both orders and the normalizing constants. Each expression in the next table integrates the kernel xy; multiplying by c makes the result 1.

Model	$dy\,dx$ order	$dx\,dy$ order	c
A	$\int_0^1\int_0^1 xy\,dy\,dx$	$\int_0^1\int_0^1 xy\,dx\,dy$	4
B	$\int_0^1\int_x^1 xy\,dy\,dx$	$\int_0^1\int_0^y xy\,dx\,dy$	8
C, E	$\int_0^1\int_0^x xy\,dy\,dx$	$\int_0^1\int_y^1 xy\,dx\,dy$	8
D	$\int_0^1\int_0^{1-x} xy\,dy\,dx$	$\int_0^1\int_0^{1-y} xy\,dx\,dy$	24

The inner integrals give the following normalization calculations:

$$\text{A: } 1 = \frac{c}{2}\int_0^1 x\,dx = \frac{c}{4}, \quad \text{B: } 1 = \frac{c}{2}\int_0^1 (x - x^3)\,dx = \frac{c}{8},$$
$$\text{C, E: } 1 = \frac{c}{2}\int_0^1 x^3\,dx = \frac{c}{8}, \quad \text{D: } 1 = \frac{c}{2}\int_0^1 x(1-x)^2\,dx = \frac{c}{24}.$$

Marginals. In every row, the displayed formula applies for $0 < x < 1$ or $0 < y < 1$, respectively; the marginal is **zero elsewhere**.

Model	$f_X(x)$: integrate out y	$f_Y(y)$: integrate out x
A	$\int_0^1 4xy\,dy = 2x$	$\int_0^1 4xy\,dx = 2y$
B	$\int_x^1 8xy\,dy = 4x(1-x^2)$	$\int_0^y 8xy\,dx = 4y^3$
C, E	$\int_0^x 8xy\,dy = 4x^3$	$\int_y^1 8xy\,dx = 4y(1-y^2)$
D	$\int_0^{1-x} 24xy\,dy = 12x(1-x)^2$	$\int_0^{1-y} 24xy\,dx = 12y(1-y)^2$

Optional practice: worked answers

Probability calculations and checks

Marginal normalization and $P(X < 1/2)$. Use an antiderivative of f_X on $(0, 1)$. Each antiderivative below equals 0 at 0 and 1 at 1, verifying normalization. Evaluating it at $1/2$ gives the requested probability.

$$\begin{aligned}
\text{A: } P(X < 1/2) &= \int_0^{1/2} 2x\,dx = [x^2]_0^{1/2} = \frac{1}{4},\\
\text{B: } P(X < 1/2) &= \int_0^{1/2} 4x(1-x^2)\,dx = [2x^2 - x^4]_0^{1/2} = \frac{7}{16},\\
\text{C, E: } P(X < 1/2) &= \int_0^{1/2} 4x^3\,dx = [x^4]_0^{1/2} = \frac{1}{16},\\
\text{D: } P(X < 1/2) &= \int_0^{1/2} 12x(1-x)^2\,dx = [6x^2 - 8x^3 + 3x^4]_0^{1/2} = \frac{11}{16}.
\end{aligned}$$

For f_Y, the same antiderivatives apply with y in place of x, exchanging the B and C rows. Thus each f_Y also integrates to 1. An area ratio is not justified here: the joint density cxy is not constant on the support.

Further practice: intersect with $x + y < 1$. The density has already been normalized; keep the same c. Integrate only over the part of the support below $y = 1 - x$.

$$\begin{aligned}
\text{A: } P(X+Y<1) &= \int_0^1 \int_0^{1-x} 4xy\,dy\,dx\\
&= 2\int_0^1 x(1-x)^2\,dx = \frac{1}{6}.\\
\text{B: } P(X+Y<1) &= \int_0^{1/2} \int_x^{1-x} 8xy\,dy\,dx\\
&= 4\int_0^{1/2} x(1-2x)\,dx = [2x^2 - \tfrac{8}{3}x^3]_0^{1/2} = \frac{1}{6}.\\
\text{C, E: } P(X+Y<1) &= \int_0^{1/2} \int_y^{1-y} 8xy\,dx\,dy\\
&= 4\int_0^{1/2} y(1-2y)\,dy = \frac{1}{6}.\\
\text{D: } P(X+Y<1) &= 1 \qquad \text{(the event contains the whole support).}
\end{aligned}$$

In B, a vertical slice runs from $y = x$ to $y = 1 - x$; these meet at $x = 1/2$. In C and E, horizontal slices give the analogous limits without splitting. If vertical slices are used for C or E instead, the upper boundary changes at $x = 1/2$:

$$\int_0^{1/2} \int_0^{x} 8xy\,dy\,dx + \int_{1/2}^1 \int_0^{1-x} 8xy\,dy\,dx = \frac{1}{6}.$$

The equal $1/6$ probabilities for A, B, C, and E do not make them the same distribution: their marginal densities and $P(X < 1/2)$ values differ.

Independence. In A, $f_X(x)f_Y(y) = 2x \cdot 2y = 4xy$ on the square, and both sides are zero outside it. Hence X and Y are independent. In B, C, D, and E, both marginals are positive throughout $(0, 1)$, but some pairs in the square lie outside the support and have joint density zero. For example, $(3/4, 1/4)$ is outside B; $(1/4, 3/4)$ is outside C and E; and $(3/4, 3/4)$ is outside D. The product of the marginals is positive at each corresponding pair, so these models are dependent. The factor xy alone does not establish independence.